STAT 400 Homework 2: Probability Functions

1) Suppose an experiment consists of tossing a two-sided coin three times.

   a) What is the size of the sample space?

$$2^3 = \mathbf{8}$$

   b) Identify the following events as subsets of the sample space:

   - E: At least two heads.

$$\{HHH, HHT, HTH, THH\}$$

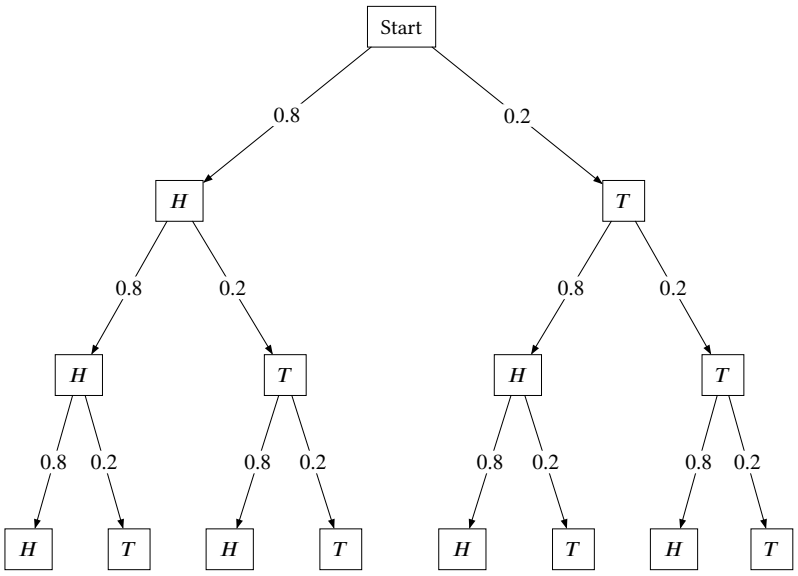
   - F: The last toss is a head.

$$\{HHH, HTH, THH, TTH\}$$

   - G: Even number of heads.

$$\{HHT, HTH, THH, TTT\}$$

   c) Suppose the tossed coin is unfair with $P(H) = 0.8$. Draw the labelled tree diagram associated to this experiment. What is the probability distribution on the sample space?



$$P(HHH) = 0.8 \cdot 0.8 \cdot 0.8 = \mathbf{0.512}$$
$$P(HHT) = 0.8 \cdot 0.8 \cdot 0.2 = \mathbf{0.128}$$
$$P(HTH) = 0.8 \cdot 0.2 \cdot 0.8 = \mathbf{0.128}$$
$$P(HTT) = 0.8 \cdot 0.2 \cdot 0.2 = \mathbf{0.032}$$
$$P(THH) = 0.2 \cdot 0.8 \cdot 0.8 = \mathbf{0.128}$$
$$P(THT) = 0.2 \cdot 0.8 \cdot 0.2 = \mathbf{0.032}$$
$$P(TTH) = 0.2 \cdot 0.2 \cdot 0.8 = \mathbf{0.032}$$
$$P(TTT) = 0.2 \cdot 0.2 \cdot 0.2 = \mathbf{0.008}$$

1) d) Use the above probability distribution to calculate the probabilities of the events E, F, and G.

$$P(E) = P(HHH) + P(HHT) + P(HTH) + P(THH)$$
$$P(E) = 0.512 + 0.128 + 0.128 + 0.128$$
$$P(E) = \mathbf{0.896}$$

$$P(F) = P(HHH) + P(HTH) + P(THH) + P(TTH)$$
$$P(F) = 0.512 + 0.128 + 0.128 + 0.032$$
$$P(F) = \mathbf{0.8}$$

$$P(G) = P(HHT) + P(HTH) + P(THH) + P(TTT)$$
$$P(G) = 0.128 + 0.128 + 0.128 + 0.008$$
$$P(G) = \mathbf{0.392}$$

2) Suppose an experiment consists of rolling a six-sided die two times.

a) What is the size of the sample space?

$$6^2 = \mathbf{36}$$

b) Identify the following events as subsets of the sample space:

- E: Sum of the two rolls is between 3 and 5 (inclusive)

$$\{(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),(3,1),(3,2),(4,1)\}$$

- F: Max of the two rolls is less than 3.

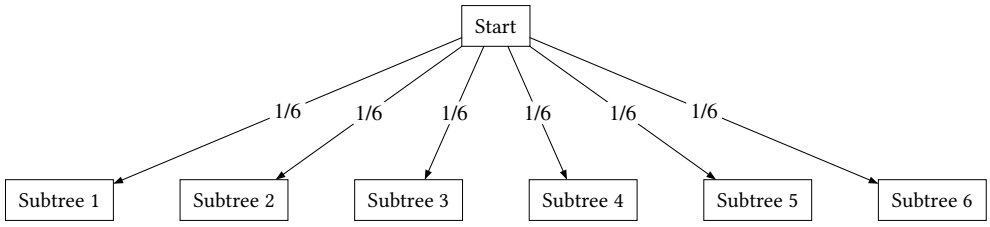
$$\{(1,1),(1,2),(2,1),(2,2)\}$$

- G: The second roll was even.

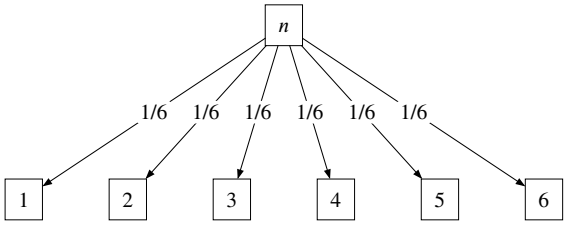
$$\{(1,2),(1,4),(1,6),(2,2),(2,4),(2,6),(3,2),(3,4),(3,6),$$
$$(4,2),(4,4),(4,6),(5,2),(5,4),(5,6),(6,2),(6,4),(6,6)\}$$

c) Suppose the rolled die is fair with $P(1) = P(2) = \cdots = P(6) = \frac{1}{6}$. Draw the labelled tree diagram associated to this experiment. What is the probability distribution on the sample space?

Main Tree:



Subtree n:



Let A be any event in the sample space.

$$P(A) = \mathbf{\frac{1}{36}}$$

2) d) Use the above probability distribution to calculate the probabilities of the events E, F, and G.

$$P(E) = P\big((1,2)\big) + P\big((1,3)\big) + P\big((1,4)\big) + P\big((2,1)\big) + P\big((2,2)\big) + P\big((2,3)\big) + P\big((3,1)\big) + P\big((3,2)\big) + P\big((4,1)\big)$$

$$P(E) = 9 \cdot \frac{1}{36}$$

$$P(E) = \mathbf{\frac{1}{4}}$$

$$P(F) = P\big((1,1)\big) + P\big((1,2)\big) + P\big((2,1)\big) + P\big((2,2)\big)$$

$$P(F) = 4 \cdot \frac{1}{36}$$

$$P(F) = \mathbf{\frac{1}{9}}$$

$$P(G) = P\big((1,2)\big) + P\big((1,4)\big) + P\big((1,6)\big) + P\big((2,2)\big) + P\big((2,4)\big) + P\big((2,6)\big) + P\big((3,2)\big) + P\big((3,4)\big) + P\big((3,6)\big) + P\big((4,2)\big) + P\big((4,4)\big) + P\big((4,6)\big) + P\big((5,2)\big) + P\big((5,4)\big) + P\big((5,6)\big) + P\big((6,2)\big) + P\big((6,4)\big) + P\big((6,6)\big)$$

$$P(G) = 18 \cdot \frac{1}{36}$$

$$P(G) = \mathbf{\frac{1}{2}}$$

3) Suppose an experiment consists of tossing a two sided coin, followed by rolling a six-sided die two times.

   a) What is the size of the sample space?

$$2 \cdot 6^2 = \mathbf{72}$$

   b) Identify the following events as subsets of the sample space:

   - E: Sum of the two rolls is between 3 and 5 (inclusive)

$$\{(H,1,2),(H,1,3),(H,1,4),(H,2,1),(H,2,2),(H,2,3),(H,3,1),(H,3,2),(H,4,1),$$
$$(T,1,2),(T,1,3),(T,1,4),(T,2,1),(T,2,2),(T,2,3),(T,3,1),(T,3,2),(T,4,1)\}$$

   - F: Max of the two rolls is less than 3.

$$\{(H,1,1),(H,1,2),(H,2,1),(H,2,2),(T,1,1),(T,1,2),(T,2,1),(T,2,2)\}$$

   - G: The second roll was even and the coin lands H.

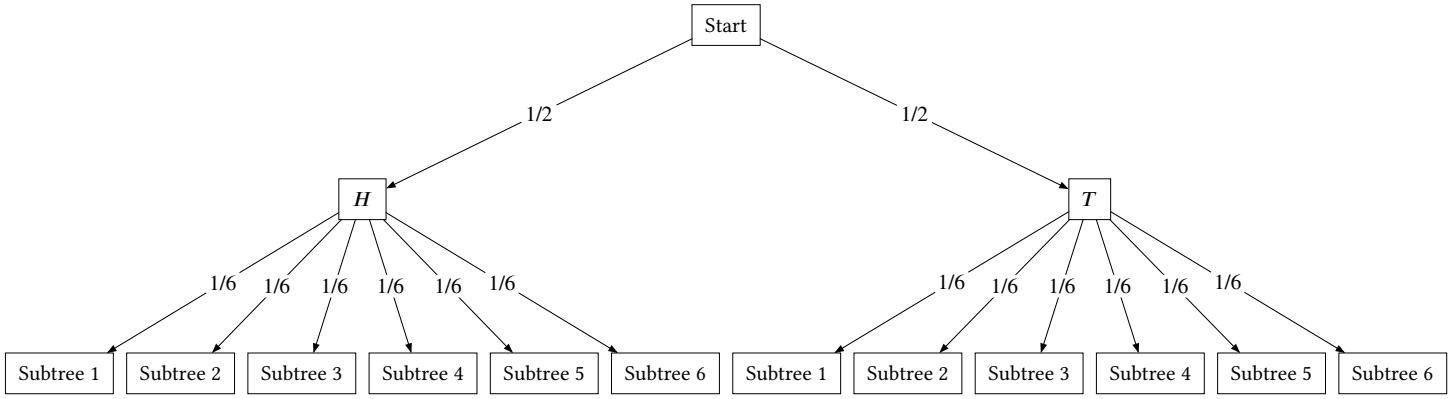
$$\{(H,1,2),(H,1,4),(H,1,6),(H,2,2),(H,2,4),(H,2,6),(H,3,2),(H,3,4),(H,3,6),$$
$$(H,4,2),(H,4,4),(H,4,6),(H,5,2),(H,5,4),(H,5,6),(H,6,2),(H,6,4),(H,6,6)$$

   - H: The coin lands H.

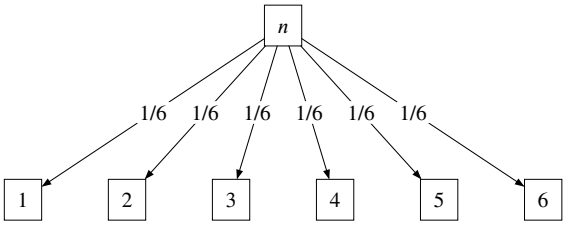
$$\{(H,1,1),(H,1,2),(H,1,3),(H,1,4),(H,1,5),(H,1,6)$$
$$(H,2,1),(H,2,2),(H,2,3),(H,2,4),(H,2,5),(H,2,6)$$
$$(H,3,1),(H,3,2),(H,3,3),(H,3,4),(H,3,5),(H,3,6)$$
$$(H,4,1),(H,4,2),(H,4,3),(H,4,4),(H,4,5),(H,4,6)$$
$$(H,5,1),(H,5,2),(H,5,3),(H,5,4),(H,5,5),(H,5,6)$$
$$(H,6,1),(H,6,2),(H,6,3),(H,6,4),(H,6,5),(H,6,6)\}$$

3) c) Suppose the tossed coin was a fair coin with $P(H) = 0.5$ and the rolled die is also fair with $P(1) = P(2) = \cdots = P(6) = \frac{1}{6}$. Draw the labeled tree diagram associated to this experiment. What is the probability distribution on the sample space?

Main Tree:



Subtree n:



Let B be any event in the sample space.

$$P(B) = \mathbf{\frac{1}{72}}$$

d) Use the above probability distribution to calculate the probabilities of the events E, F, G, and H.

$$\begin{aligned} P(E) = {} & P\big((H,1,2)\big) + P\big((H,1,3)\big) + P\big((H,1,4)\big) + P\big((H,2,1)\big) \\ & + P\big((H,2,2)\big) + P\big((H,2,3)\big) + P\big((H,3,1)\big) + P\big((H,3,2)\big) + P\big((H,4,1)\big) \\ & + P\big((T,1,2)\big) + P\big((T,1,3)\big) + P\big((T,1,4)\big) + P\big((T,2,1)\big) \\ & + P\big((T,2,2)\big) + P\big((T,2,3)\big) + P\big((T,3,1)\big) + P\big((T,3,2)\big) + P\big((T,4,1)\big) \end{aligned}$$

$$P(E) = 18 \cdot \frac{1}{72}$$

$$P(E) = \mathbf{\frac{1}{4}}$$

3) d)

$$P(F) = P\big((H,1,1)\big) + P\big((H,1,2)\big) + P\big((H,2,1)\big) + P\big((H,2,2)\big)$$
$$+P\big((T,1,1)\big) + P\big((T,1,2)\big) + P\big((T,2,1)\big) + P\big((T,2,2)\big)$$

$$P(F) = 8 \cdot \frac{1}{72}$$

$$P(F) = \mathbf{\frac{1}{9}}$$

$$P(G) = P\big((H,1,2)\big) + P\big((H,1,4)\big) + P\big((H,1,6)\big)$$
$$+P\big((H,2,2)\big) + P\big((H,2,4)\big) + P\big((H,2,6)\big)$$
$$+P\big((H,3,2)\big) + P\big((H,3,4)\big) + P\big((H,3,6)\big)$$
$$+P\big((H,4,2)\big) + P\big((H,4,4)\big) + P\big((H,4,6)\big)$$
$$+P\big((H,5,2)\big) + P\big((H,5,4)\big) + P\big((H,5,6)\big)$$
$$+P\big((H,6,2)\big) + P\big((H,6,4)\big) + P\big((H,6,6)\big)$$

$$P(G) = 18 \cdot \frac{1}{72}$$

$$P(G) = \mathbf{\frac{1}{4}}$$

$$P(H) = P\big((H,1,1)\big) + P\big((H,1,2)\big) + P\big((H,1,3)\big) + P\big((H,1,4)\big) + P\big((H,1,5)\big) + P\big((H,1,6)\big)$$
$$+P\big((H,2,1)\big) + P\big((H,2,2)\big) + P\big((H,2,3)\big) + P\big((H,2,4)\big) + P\big((H,2,5)\big) + P\big((H,2,6)\big)$$
$$+P\big((H,3,1)\big) + P\big((H,3,2)\big) + P\big((H,3,3)\big) + P\big((H,3,4)\big) + P\big((H,3,5)\big) + P\big((H,3,6)\big)$$
$$+P\big((H,4,1)\big) + P\big((H,4,2)\big) + P\big((H,4,3)\big) + P\big((H,4,4)\big) + P\big((H,4,5)\big) + P\big((H,4,6)\big)$$
$$+P\big((H,5,1)\big) + P\big((H,5,2)\big) + P\big((H,5,3)\big) + P\big((H,5,4)\big) + P\big((H,5,5)\big) + P\big((H,5,6)\big)$$
$$+P\big((H,6,1)\big) + P\big((H,6,2)\big) + P\big((H,6,3)\big) + P\big((H,6,4)\big) + P\big((H,6,5)\big) + P\big((H,6,6)\big)$$

$$P(H) = 36 \cdot \frac{1}{72}$$

$$P(H) = \mathbf{\frac{1}{2}}$$

4) Suppose we toss a two-sided coin until H shows up.

a) Identify the following events as subsets of the sample space:

- E: Even number of tails before the first H.

$$\{H, TTH, TTTTH, TTTTTTH, \dots\}$$

- F: At least 5 tails before the first H.

$$\{TTTTTH, TTTTTTH, TTTTTTTH, \dots\}$$

- G: At most 4 tails before the first H.

$$\{H, TH, TTH, TTTH, TTTTH\}$$

b) Suppose the coin is tossed such that $P(H) = p \in (0, 1)$. Use this information to calculate the probabilities of the events E, F, and G.

$$\begin{aligned}
P(E) &= P(TTH) + P(TTTTH) + P(TTTTTTH) + \cdots \\
P(E) &= (1-p)^2 p + (1-p)^4 p + (1-p)^6 p + \cdots \\
P(E) &= p\left((1-p)^2 + (1-p)^4 + (1-p)^6 + \cdots\right) \\
P(E) &= p \sum_{n=0}^{\infty} (1-p)^{2n} \\
P(E) &= \frac{p}{1-(1-p)^2} \\
P(E) &= \boldsymbol{\frac{1}{2-p}}
\end{aligned}$$

$$\begin{aligned}
P(F) &= P(TTTTTH) + P(TTTTTTH) + P(TTTTTTTH) + \cdots \\
P(F) &= (1-p)^5 p + (1-p)^6 p + (1-p)^7 p + \cdots \\
P(F) &= (1-p)^5 p\left(1 + (1-p) + (1-p)^2 + \cdots\right) \\
P(F) &= p(1-p)^5 \sum_{n=0}^{\infty} (1-p)^n \\
P(F) &= \frac{p(1-p)^5}{1-(1-p)} = \boldsymbol{(1-p)^5}
\end{aligned}$$

$$\begin{aligned}
P(G) &= P(H) + P(TH) + P(TTH) + P(TTTH) + P(TTTTH) \\
P(G) &= p + p(1-p) + p(1-p)(1-p) + p(1-p)(1-p)(1-p) + p(1-p)(1-p)(1-p)(1-p) \\
P(G) &= p\left(1 + (1-p) + (1-p)^2 + (1-p)^3 + (1-p)^4\right) \\
P(G) &= p \sum_{n=0}^{4} (1-p)^n \\
P(G) &= p \cdot \frac{1-(1-p)^5}{1-(1-p)} \\
P(G) &= \boldsymbol{1-(1-p)^5}
\end{aligned}$$